

2(a)	$\rho = m / V$ or $\rho = m / Ah$	B1
	$p = F / A$ or $p = W / A$	B1
	$p = [\rho Ahg] / A$ or $p = [\rho Vg] / [V / h]$ (so) $p = \rho gh$	A1

Question	Answer	Marks
2(b)(i)	1. weight/gravitational (force) upthrust (force)/buoyancy (force) drag/viscous/frictional (force)/fluid resistance/resistance	B1
	2. weight = upthrust + viscous (force)	B1
2(b)(ii)	- decrease in (gravitational) potential energy (of sphere) due to decrease in height (since $E_p = mgh$) - increase in thermal energy due to work done against viscous force/drag - loss/change of (total) E_p equal to gain/change in thermal energy <i>Any 2 points.</i>	B2
2(c)(i)	atmospheric pressure = $9.1(0) \times 10^4$ Pa	A1
2(c)(ii)	$(\Delta)p = \rho g(\Delta)h$ $(9.15 - 9.10) \times 10^4 = \rho \times 9.81 \times (0.17 - 0.10)$	C1
	$\rho = 730$ (728) kg m^{-3}	A1

Question	Answer	Marks
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